

Daffodil International University
Faculty of Science & Information Technology
Department of Computer Science & Engineering

Quiz Examination

Course Code: CSE321 Course Title: System Analysis and Design

Level: 3 Term: 2 | Batch: 67 | Section: RE_a(3c)

Time: 40 Minutes	Total Marks: 15
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Answer ALL Questions. Each question carries 5 marks.

Question 1: [Topics: Managing the IS Project & Project Scheduling]

[Marks: 2 + 1 + 2 = 5]

- (a) Define the following terms with examples: (i) Project (ii) Project Manager (iii) Deliverable (iv) Slack Time (v) Gantt Chart.
- (b) Distinguish between a Gantt Chart and a Network Diagram. Mention TWO key features of each.
- (c) A project has the following activities with their estimated times (in weeks): Optimistic (o), Realistic (r), and Pessimistic (p). Using the PERT formula $ET = (o + 4r + p) / 6$, calculate the Expected Time (ET) for each activity and identify which activity will take the longest to complete.

Activity	Optimistic (o)	Realistic (r)	Pessimistic (p)	ET = ?
System Requirements Gathering	2	4	6	
Database Design	1	3	5	
UI Prototyping	3	5	7	
Module Coding	4	6	8	

Question 2: [Topic: Designing Effective Output – Invoice Design]

[Marks: 5]

Read the following scenario carefully and answer all parts:

Scenario: Mr. Rahim placed an online order on "ShopBD.com", a popular e-commerce platform in Bangladesh. He ordered the following products: (1) Wireless Mouse – Qty: 2, Unit Price: ₳850; (2) USB-C Hub – Qty: 1, Unit Price: ₳1,200; (3) Laptop Stand – Qty: 1, Unit Price: ₳650. The platform applies a 5% VAT on the subtotal and offers a flat ₳100 shipping charge. Mr. Rahim paid using his VISA credit card. The order was placed on 05 April 2026 with Order No. SBD-20260405 and Invoice No. INV-9901.

- (a) Using the scenario above, design a complete and well-structured Invoice for this e-commerce order. Your invoice must include the following sections:
- (i) Header section – Company name, logo placeholder, and invoice title.
 - (ii) Customer & Order Info section – Customer name, order number, invoice number, order date, and payment method.
 - (iii) Itemized Product Table – Columns for Qty, Product Description, Unit Price, and Total Amount.
 - (iv) Summary / Totals section – Subtotal, VAT (5%), Shipping Charge, and Grand Total.
 - (v) Footer section – A thank-you note and return/refund policy statement.

Question 3: [Topics: System Testing & Information Security and Cybercrime]

[Marks: 2 + 2 + 1 = 5]

- (a) Differentiate between Failure, Fault/Bug, and Error in the context of system testing. Give a suitable example for each.
- (b) Compare White Box Testing and Black Box Testing. In the table below, fill in the correct description for each property:

Property	White Box Testing	Black Box Testing
Also known as		
Knowledge of internal code required?		
Who performs it?		
Test cases derived from		

- (c) Define the following cybersecurity threats and explain how each can harm an organization:

(i) Trojan Horse (ii) Spyware